FlickerTac: Flickering LED Driven Photometric Stereo for Event Vision-Based Tactile Sensors

Mohamad Halwani, Akram Khairi, Hussain Sajwani, Eslam Sherif, Laith AbuAssi, Sajid Javed, Abdelqader Abusafieh, Abdulla Ayyad, and Yahya Zweiri

***Abstract*—Tactile sensing at high speed and resolution is critical for robotic perception and control. Existing vision-based tactile sensors (VBTSs) achieve high spatial accuracy but suffer from degraded performance in dynamic interactions due to motion blur and low frame rates, limiting their effectiveness in fast manipulation. Event-based cameras, with their high-temporal-resolution output, offer a promising alternative—but their inability to capture static scenes and absolute intensities has hindered their use for dense 3D reconstruction even under static contact conditions. We introduce FlickerTac, the first Event Vision-Based Tactile Sensor (EVBTS) capable of performing photometric stereo-based dense 3D reconstruction. FlickerTac replaces conventional 3-channel RGB illumination with time-multiplexed, phase-shifted flickering LEDs to encode directional shading into temporally separated channels. A lightweight neural network then maps these spatio-temporal representations directly to surface gradients, which are integrated to recover dense depth maps. Our design enables high-speed, accurate tactile 3D geometry estimation. Extensive experiments demonstrate that FlickerTac achieves millimeter-scale depth accuracy competitive with state-of-the-art VBTSs, while operating at 125 Hz—unmatched by existing frame-based VBTSs.**

***Index Terms*—Event-based vision, neuromorphic, photometric stereo, 3D reconstruction, vision-based tactile sensor, neuromorphic sensing.**

I. Introduction

REPLICATING the human sense of touch is critical for enhancing robotic dexterity, enabling precise manipulation, alignment, and perception in occluded environments. Vision-based tactile sensors (VBTSs) have gained significant attention in robotic perception for their ability to capture high-resolution 3D contact geometry, providing rich tactile feedback [1], [2], [3], [4], [5]. However, traditional VBTSs rely on frame-based cameras and suffer from motion blur, latency, and low sampling rates (30–60 Hz), limiting their responsiveness to rapid deformations and transient events such as incipient slip [6], [7], [8], [9]. Event-based cameras offer a promising alternative by capturing asynchronous brightness changes with high temporal resolution and low latency [10], driving the development of Event Vision-Based Tactile Sensors (EVBTSs).

Current EVBTS research focuses on slip detection [9], [11], [12], vibration sensing [12], contact pose estimation [6], [13], texture classification [14], [15], [16], [17], and force measurement [18], primarily using the Marker Displacement Method (MDM) [19]. MDM tracks marker motion on flexible skins to encode deformations into brightness changes for event-based cameras. Alternatively, reflective membrane techniques leverage structured illumination to extract shading cues for photometric stereo and dense 3D reconstruction [20], [21], [22]. Yet, all existing EVBTSs rely solely on MDM designs [6], [12], [13], [14], [15], [16], overlooking tasks that require high-resolution geometry—such as fine texture sensing, surface inspection, and in-hand manipulation. Traditional VBTSs like GelSight [21], [23], DIGIT [22], and others [24], [25], [26], [27] use RGB lighting to recover depth via photometric stereo. However, RGB-based methods require frame sensors, limiting temporal resolution and rendering them less suitable for fast dynamic contacts.

Adapting structured illumination to EVBTSs is challenging due to the inherent characteristics of event-based cameras, which detect brightness changes rather than absolute intensity. In static scenes, this results in limited data for extracting shading cues needed for photometric stereo, and even in dynamic cases, surface gradient estimation remains unreliable. Consequently, existing EVBTSs rely on marker-based elastomers and trackers for tasks like slip and vibration sensing, but these methods are inadequate for achieving dense 3D reconstruction and detailed object pose estimation. This limitation underscores the need to incorporate structured illumination, similar to photometric stereo, within the event-based tactile sensing framework.

In this paper, we propose FlickerTac, the first Event Vision-Based Tactile Sensor (EVBTS) capable of dense 3D reconstruction via photometric stereo as seen in Fig. 1. FlickerTac replaces RGB-based illumination with three phase-shifted monochromatic LEDs, using temporal modulation instead of spectral separation to generate structured shadows. These asynchronously flickering LEDs enable the generation of distinct illumination

Received 12 January 2026; accepted 6 May 2026. Date of publication 11 May 2026; date of current version 19 May 2026. This article was recommended for publication by Associate Editor Y. She and Editor A. Banerjee upon evaluation of the reviewers' comments. This work was supported in part by Sandooq Al Watan, United Arab Emirates under Grant SWARD-S22-015, STRATA Manufacturing PJSC, in part by Advanced Research and Innovation Center (ARIC), funded by Aerospace Holding Company LLC, a holly-owned subsidiary of Mubadala Investment Company PJSC, and in part by the Khalifa University of Science and Technology, United Arab Emirates. *(Corresponding author: Mohamad Halwani.)*

Mohamad Halwani and Yahya Zweiri are with Advanced Research and Innovation Center (ARIC), Abu Dhabi 127788, UAE, and also with the Department of Aerospace Engineering, Khalifa University, Abu Dhabi 127788, UAE (e-mail: mohamad.halwani@ku.ac.ae).

Akram Khairi, Hussain Sajwani, Eslam Sherif, Laith AbuAssi, and Abdulla Ayyad are with Advanced Research and Innovation Center (ARIC), Abu Dhabi 127788, UAE.

Sajid Javed is with the Department of Computer Science, Khalifa University, Abu Dhabi 127788, UAE.

Abdelqader Abusafieh is with Advanced Research and Innovation Center (ARIC), Abu Dhabi 127788, UAE, and also with the Department of Research and Development, Strata Manufacturing PJSC, Al Ain 86519, UAE.

Video: https://youtu.be/Uo7oEXmbq40.

This article has supplementary downloadable material available at https://doi.org/10.1109/LRA.2026.3692331, provided by the authors.

Digital Object Identifier 10.1109/LRA.2026.3692331

© 2026 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/



Fig. 1. FlickerTac sensor mounted on a UR10 robotic arm.

channels that are utilized for photometric stereo depth estimation. Unlike conventional photometric stereo, which relies on absolute intensity, FlickerTac accumulates events over time and separates LED contributions based on known flicker phases, forming event-based illumination channels. A learning-based model then maps local patches to surface gradients, enabling dense 3D shape recovery—even under static contact where event generation typically fails. This high-speed, markerless sensing pipeline allows robust reconstruction for fine manipulation tasks where traditional VBTSs are hindered by latency and limited dynamic response. A comparison with conventional methods is shown in Fig. 2. The key contributions of our work are summarized as follows:

- We introduce FlickerTac, the first event-based photometric stereo tactile sensor for accurate and high-speed 3D contact reconstruction.
- We present the first dense 3D reconstruction method for event vision-based tactile sensors,
  enabling accurate, fast, and high-resolution depth estimation for tactile surface reconstruction.
- We develop a time-multiplexed structured illumination method for EVBTS, replacing RGB lighting with phase-shifted monochromatic LEDs to generate independent event-based illumination channels.

II. Related Work

Event-based cameras have been explored for 3D reconstruction using structured light, multi-view stereo (EMVS), and event-based photometric stereo (EventPS). Structured light methods [28] achieve accurate depth via projected patterns, but require external illumination, precise synchronization, and are sensitive to surface properties—making them unsuitable for compact, deformable EVBTSs. EMVS [29] reconstructs depth by aggregating events from multiple viewpoints, assuming known sensor motion. This assumption fails in EVBTSs, where the sensor is stationary and surface motion dominates. Moreover, EMVS yields only semi-dense outputs, limiting tactile resolution. EventPS [30] estimates surface normals from radiance changes caused by a moving light source, which is impractical to implement within compact tactile sensors. These limitations highlight the need for a self-contained EVBTS approach using embedded, flickering LEDs to enable structured shading without external projectors or moving parts.

Event cameras' use in tactile sensing remains limited despite success in 3D reconstruction. Existing EVBTSs rely on Marker Displacement Methods (MDM) [19] due to event cameras' inability to capture absolute intensity on static reflective surfaces. Systems like Evetac [12], NeuroTac [14], and NeuTac [6] track marker motion for slip detection and contact pose estimation but cannot perform dense 3D reconstruction. Some works use graph-based networks [13] to infer contact properties, yet remain limited to dynamic interactions without recovering surface geometry. No existing EVBTS leverages structured illumination for 3D shape recovery—highlighting the need for continuous event generation and shading-based depth estimation without marker reliance.

III. Methodology

A. Sensor Design and Fabrication

The proposed FlickerTac sensor depicted in Fig. 1 is an EVBTS that introduces a novel time-multiplexed structured illumination strategy to generate dense photometric channels from asynchronous event data. It integrates an IniVation DVXplorer Mini event-based camera,[1] a structured illumination module with three phase-shifted monochromatic LEDs, and a reflective tactile skin similar to DIGIT [22]. The silicone membrane, coated with white silicone ink, provides robust directional shading cues under contact and offers better durability than marker-based alternatives [12], [22]. Unlike MDM approaches that track surface features, FlickerTac leverages retrographic sensing [20] to extract dense geometry via shading alone. All components are enclosed in a compact 3D-printed housing for seamless robotic integration.

The event-based camera asynchronously captures brightness changes, enabling low-latency, high-speed sensing—but lacking absolute intensity makes it unsuitable for standard photometric stereo under static light. FlickerTac resolves this by driving three monochromatic LEDs in phase-shifted cycles, arranged 120° apart to capture directional shading essential for photometric inference [31], similar to RGB-based systems like GelSight [21] and DIGIT [22].

FlickerTac's time-multiplexed illumination supports more than three phase-shifted LEDs, enabling additional independent illumination directions for higher-fidelity 3D reconstruction. While frame-based VBTSs can similarly use multiple sequential illuminations [32], FlickerTac achieves time-division at event-level temporal resolution (microseconds) rather than frame-level (tens of milliseconds), enabling substantially higher photometric stereo update rates.

B. Event-Based Camera Fundamentals

The core sensing element of FlickerTac is the IniVation DVXplorer Mini event-based camera (temporal resolution: 200 μs, spatial resolution: 640 $\times$ 480 pixels, throughput: 450 MEPS). Event sensors generate $e_i = (x_i, y_i, t_i, p_i)$ when log light intensity $L(x, y, t)$ changes beyond threshold $\pm C$, eliminating motion blur which is ideal for rapid tactile deformations.

Despite this asynchronous architecture, the DVXplorer Mini uses a synchronous on-chip read-out, batching events within configurable intervals under a shared timestamp[2]. Under high event rates, this batching introduces latency and temporal ambiguity, complicating timing-sensitive measurements. Furthermore, event-based cameras do not report absolute intensity,

[1] https://inivation.com/wp-content/uploads/2023/03/DVXplorer-Mini.pdf
[2] https://docs.inivation.com/software/dv/gui/adjust-camera-settings.html

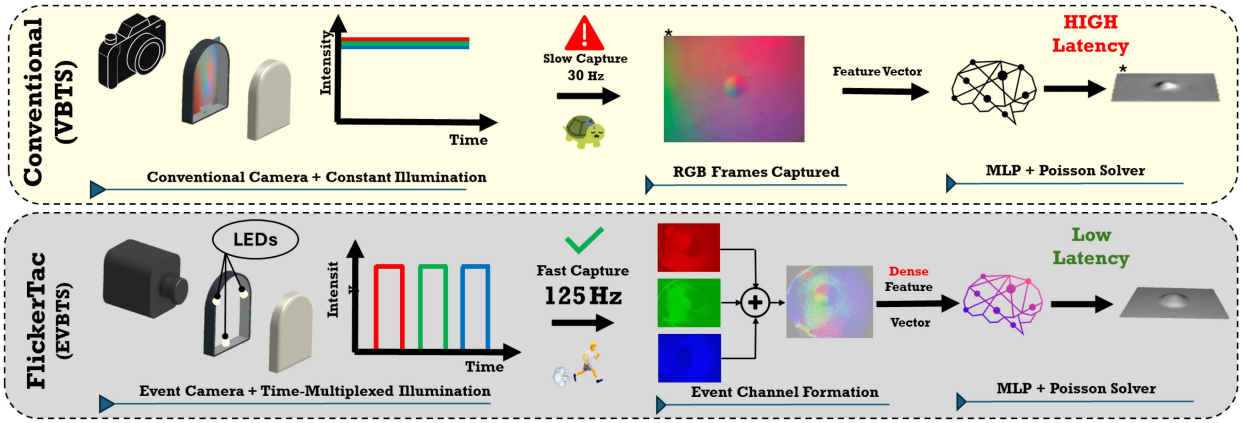


Fig. 2. Comparison of conventional VBTS and FlickerTac pipelines. Conventional VBTS uses RGB frames, while FlickerTac uses time-multiplexed LED illumination and event channels for 125 Hz 3D reconstruction. *Images adapted from [23].

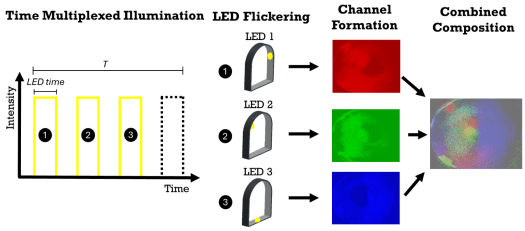


Fig. 3. Time-multiplexed illumination. Three LEDs sequentially generate directional event channels; colors are for visualization only.

limiting compatibility with classic photometric stereo methods. FlickerTac overcomes these challenges by driving three monochromatic LEDs at phase-shifted flicker frequencies (see Section III-C). This produces structured event streams that encode directional shading cues in the temporal domain, enabling accurate depth reconstruction while operating within the DVXplorer's read-out constraints.

C. Time-Multiplexed Illumination and Channels Formation

1) Phase-Shifted LED Illumination Strategy: To enable EVBTS photometric stereo, FlickerTac utilizes a structured illumination strategy controlled by a microcontroller. The illumination setup comprises three monochromatic LEDs. Each LED is modulated using precise square-wave signals generated by the microcontroller at a fixed flickering frequency, employing distinct temporal phase shifts.

The LEDs follow a time-multiplexed sequence with each LED active for $T/8$ of the cycle period $T = 8 \times LED_time$. This division into eight segments introduces idle intervals between LED activations, reducing inter-channel ambiguity despite the camera's timestamp quantization (Fig. 3). FlickerTac's time-multiplexed approach supports scalable multi-LED photometric stereo at event-level temporal resolution.

The LED illumination is modeled by a binary gate

$$I_k(t) = \mathbf{1}_{\left[t_k^{\text{on}}, t_k^{\text{off}}\right]}\left(t \bmod T\right), \tag{1}$$

where t is absolute time, T is the flicker-cycle period, and $t_k^{\text{on}}/t_k^{\text{off}}$ are the start/end times (within $[0, T]$) of LED k's. The indicator $\mathbf{1}_{[a,b]}(u)$ equals 1 if $u \in [a, b]$ and 0 otherwise, so $I_k(t) \in \{0, 1\}$ gates LED k ON/OFF within each cycle. The modulo $(t \bmod T) \in [0, T]$ maps absolute time into the current cycle, ensuring only one LED is active at the current cycle.

This time-division approach guarantees clear temporal separation between the LEDs, enabling unambiguous assignment of events to the correct illumination source. Practically, it is implemented using digital output pins on the microcontroller, which switch between HIGH (ON) and LOW (OFF) based on these fixed timing intervals.

2) Event Channel Construction and Demultiplexing: The DVXplorer Mini records polarity events generated by the phase-shifted LED sequence and groups all events occurring in the same global-shutter interval into a single "event packet" sharing the same timestamp t. Let $e_i = (x_i, y_i, t_i, p_i)$ be the i-th event and let T be the LED flicker period and $\mathcal{I}_k = [t_k^{\text{on}}, t_k^{\text{off}}]$ the slot for LED k. We demultiplex by time-gating and accumulate both polarities (magnitude) to form directional channels C:

$$C_k(x, y) = \sum_{\substack{e_i, \\ t_i \bmod T \in \left[t_k^{\text{on}} - \epsilon,\, t_k^{\text{off}} + \epsilon\right]}} |p_i|, \qquad k \in \{1, 2, 3\}, \tag{2}$$

where events are accumulated over the entire slot (rising and falling transitions), with a short guard intervals ϵ between slots to suppress inter-channel leakage.

DVXplorer pixels emit events asynchronously, but the chip uses synchronous readout. The FIFO buffer can drain $\sim 90,000$ events in $< 400\,\mu$s [33], with sustained bandwidth up to 450 MEPS limiting overall throughput.

Our tactile application requires (i) the *largest possible Region of Interest ROI* to maximize spatial resolution and (ii) a *high update rate* to capture fast contact dynamics. We tailor the LED timing so that the instantaneous event burst produced by a single LED never approaches the maximum readout speed of 450 MEPS while keeping high update rate.

Table I shows that at a 3D reconstruction update rate of 125 Hz ($T = 8$ ms) each LED window lasts 1 ms and can accommodate up to 450 k events, sufficient to fully utilize the DVXplorer Mini's 307 k pixel resolution. This setting therefore (a) avoids buffer congestion, (b) preserves per-pixel temporal ordering, and (c) prevents inter-channel cross-talk, while exceeding the 30–60 Hz of frame-based VBTSs and providing the temporal resolution needed for rapid contact dynamics where local gripper control loops benefit from high frequency feedback [34].

TABLE I
THROUGHPUT BUDGET AS A FUNCTION OF THE DESIRED PHOTOMETRIC STEREO UPDATE RATE ($f_{\text{PS}} = 1/T$). THE MAXIMUM NUMBER OF EVENTS THAT CAN BE DRAINED WITHIN ONE LED WINDOW IS $N_{\max} = 450\,\text{MEPS} \times \texttt{LED_time}$.

f_{PS} (Hz)	LED window freq. (Hz)	LED_time (ms)	$N_{\max}$ events
1000	8000	0.125	56 k
500	4000	0.250	113 k
250	2000	0.500	225 k
125	**1000**	**1.000**	**450 k**
100	800	1.250	563 k

During data collection, the camera is set to its highest event-frames-per-second (eFPS) bias and a low contrast threshold. The former keeps the output stage safely within the sustained-bandwidth envelope, whereas the latter suppresses shot noise and electrical jitter, further improving channel separation.

By marrying conservative time division, quantitative bandwidth analysis and noise-aware biasing, FlickerTac acquires three clean, event channels that retain the shading cues required for reliable photometric stereo as seen in Fig. 3, even under the DvXplorer Mini synchronous read-out regime and limited temporal precision.

D. 3D Reconstruction

Photometric stereo estimates 3D shape by observing a surface under multiple known lighting directions [31]. Assuming Lambertian reflectance, the radiance at pixel $\mathbf{x} = (x, y)$ with normal $\mathbf{n}_x$ and albedo a_x, under lighting direction $\mathbf{L}(t)$, is:

$$\hat{I}_x(t) = \max\left[0,\, a_x\,(\mathbf{n}_x \cdot \mathbf{L}(t))\right]. \tag{3}$$

Recovering the scaled normal $\mathbf{g}_x = a_x\,\mathbf{n}_x$ requires at least three linearly independent light sources [35]. In FlickerTac, the three phase-shifted LEDs act as such sources, analogous to RGB photometric stereo. Despite minor non-Lambertian effects (e.g., gloss or scattering), this assumption remains standard in VBTS applications [21], [23], and our data-driven model implicitly compensates for deviations by calibrating on silicone membranes.

After decoding the event stream into aligned channels (C_1, C_2, C_3) (see Section III-C), each pixel's shading response is mapped to surface gradients using a Multi-Layer Perceptron (MLP). For each pixel, we extract 3×3 patches from all channels, plus coordinates, forming a 29-D feature vector:

$$\mathbf{v}_{x,y} = [C_{1_{3\times3}},\, C_{2_{3\times3}},\, C_{3_{3\times3}},\, x,\, y]\,. \tag{4}$$

The 3×3 neighborhood improves gradient estimation stability and suppresses noise in the sparse event stream. This vector feeds a lightweight MLP (29–64–64–2), outputting (G_x, G_y) surface gradients.

To train the MLP, a 4 mm radius sphere is indented into the sensor at known depths (Section III-E). Contact regions are labeled on reference frames, and surface gradients are analytically derived from the known sphere geometry, using a calibrated pixel-to-millimeter factor f_{pix}. Each training pair consists of a feature vector $\mathbf{v}_{x,y}$ and its corresponding ground-truth gradients $(G_{x,\text{gt}}, G_{y,\text{gt}})$. The MLP is trained using a mean squared error (MSE) loss between the predicted and ground-truth gradients:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{2}\left[\left(\hat{G}_x - G_{x,\text{gt}}\right)^2 + \left(\hat{G}_y - G_{y,\text{gt}}\right)^2\right]. \tag{5}$$

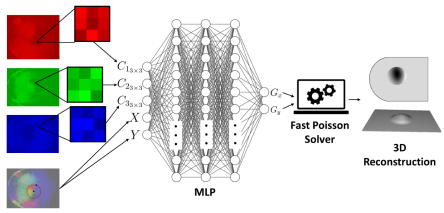


Fig. 4. 3D Reconstruction Pipeline. Patch-based features from channels are used by an MLP to regress surface gradients to recover depth.

We minimize the average $\mathcal{L}_{\text{MSE}}$ over all training samples using Adam optimizer with a learning rate of 0.001.

At inference time, the MLP delivers two gradient images G_x, G_y. We recover the height map $Z(x, y)$ by solving the discrete Poisson equation:

$$\nabla^2 Z = \frac{\partial G_x}{\partial x} + \frac{\partial G_y}{\partial y}, \tag{6}$$

using a fast DCT-based solver that enforces global integrability while retaining real-time performance, similar to the method used in [23]. The full pipeline is depicted in Fig. 4.

E. Dataset and Calibration

Data were collected with the FlickerTac sensor mounted on a UR10 robot, indenting a fixed 4 mm radius steel sphere at varied positions across the tactile skin to ensure spatial coverage and at arbitrary depths within the safe deformation range, resulting in 40 distinct press events. For each indentation, the raw event stream was recorded and later decomposed into three illumination channels C_1, C_2, C_3, as seen in Section III-C.

A cumulative event image was generated per indentation, and the circular contact patch was manually labeled. A single calibrated press using vernier calipers yielded the pixel-to-millimeter scale f_{pix}. Given the patch center (x_c, y_c), radius r_{pix}, and known sphere radius $R_b = 4$ mm, ground-truth surface gradients $(G_{x,\text{gt}}, G_{y,\text{gt}})$ were computed analytically for each pixel in contact; others were set to zero.

We used 32 indentations for training and 8 for testing (80/20 split). Training samples were extracted from all pixels with full 3×3 neighborhoods inside the labeled region. To enhance generalization, 10% of background pixels (meeting the same constraint) were also included, ensuring the MLP learns both surface gradients in contact regions and flat responses elsewhere.

IV. EXPERIMENTS

A. Experimental Setup

To evaluate the 3D reconstruction performance of the FlickerTac sensor, experiments were conducted using a UR10 robotic manipulator to perform precise, repeatable indentations against static targets of known geometry. As shown in Fig. 1, the FlickerTac sensor was mounted on the UR10's end-effector. The robot executed vertical indentations into various test objects positioned beneath the sensor on a flat surface, allowing controlled contact at specific lateral positions and depths. The measured thickness of the tactile skin was approximately 4.29 mm.

A calibrated steel sphere with a radius of 4 mm was used as the primary reference object for ground truth gradient derivation. The sphere was chosen for its smooth and continuous

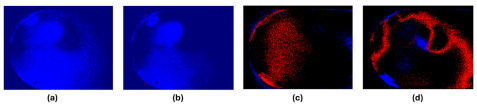


Fig. 5. LED evaluation. (a) Uniform directional illumination under optimal biasing. (b) Poor illumination due to low LED power or partial skin detachment. Examples of FIFO buffer congestion and inter-channel cross-talk at 200 eFPS: (c) and (d) show representative leakage patterns.

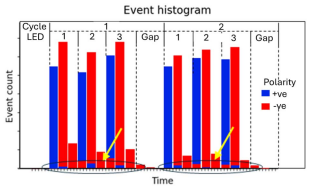


Fig. 6. LED-window histogram showing minor inter-channel leakage

curvature, which enables accurate analytical computation of surface gradients for training and evaluating the MLP model. During each indentation, the structured event stream produced by the phase-shifted LED flickering was recorded, with the LED timing carefully configured to stay within the read-out bandwidth constraints of the event-based camera, as discussed earlier in Section III-C.

In addition to the calibration sphere, three 3D-printed objects were used to evaluate spatial resolution and sensitivity to depth discontinuities: a hexagonal platform (1.5 mm height, 0.5 mm indentation), a bumps profile (1.50 mm/0.75 mm max/min heights), and concentric circular steps (0.6 mm/1.2 mm heights). All objects were printed at $\leq$0.1 mm layer height using a Creality K1C printer to ensure sub-millimeter fidelity. Experiments were conducted under static contact conditions, demonstrating FlickerTac's ability to recover surface geometry without relying on relative motion—an outcome enabled by continuous LED flickering, which generates structured shading cues even when without motion.

B. LED and Event-Based Camera Evaluation

To ensure robust depth estimation from structured illumination, we evaluated the influence of LED output quality and event-based camera performance, specifically the event throughput (eFPS) and demultiplexing reliability on the quality of the three directional event channels.

Fig. 5(a) illustrates an event channel captured under optimal conditions, where a properly biased LED produces uniform directional illumination and sufficient contrast across the contact surface. In contrast, Fig. 5(b) shows a degraded illumination example due to a low-powered LED or partial detachment of the reflective skin. This leads to insufficient shading contrast, resulting in sparsely triggered events and weakened channel definition. Ensuring balanced current delivery and symmetric LED placement is therefore essential to preserve shading integrity across channels.

Although each illumination slot is temporally isolated, slight background color variations may appear in the visualized event channels. It is important to note that these colors represent a false-color visualization—each demultiplexed channel (C_1, C_2, C_3) is assigned red, green, and blue hues for display, and the underlying event data remain binary per channel. The observed color shifts arise from intrinsic physical effects, not synchronization errors. The primary cause is LED phosphor decay: when an LED switches off, residual emission is detected by the event camera as OFF events extending beyond the active window and potentially persisting as inconsistent residuals in the subsequent window, as seen in the histogram in Fig. 6. Furthermore, the high sensitivity of the event camera may trigger minor positive event trails at the start of an illumination window as pixels cross their activation thresholds; however, this electronic artifact is minimal compared to the phosphor-driven negative trails. Internal reflections and the partial translucency of the tactile skin contribute additional scattered light. Because these leakage sources vary temporally and spatially, they manifest as inconsistent background colors across frames. To mitigate these effects, we introduced temporal guard intervals (Section III-C2), neighborhood-based (3×3) encoding and 10% background training pixels samples (Section III-D).

Beyond LED performance, the temporal resolution and event throughput of the DVXplorer Mini directly influence the clarity of channel separation. Because FlickerTac encodes directional lighting through phase-shifted LED flickering, each illumination cycle must remain within the camera's sustained bandwidth limits. When operated under high readout speed settings (10000 eFPS), we observed minimal event overlap between channels. Fig. 3 demonstrates this clean separation: each LED's active window produces a concentrated burst of events confined to its expected timing interval. However, at lower readout speed settings, such as 200 eFPS, the read-out bottleneck begins to degrade channel purity. Fig. 5 c and Fig. 5 d shows two examples of a significant inter-channel leakage, with residual events from one LED spilling into adjacent windows. This cross-talk arises from on-chip FIFO buffer congestion and the shared timestamping of batched events under heavy load.

These results validate the choice of a 125 Hz photometric stereo update rate, which corresponds to a full LED cycle time of 8 ms with 1 ms per LED. As shown in Table I, this configuration supports a maximum of 450 k events per LED window, safely avoids FIFO overflow. In addition, these evaluations confirm that the event-based camera parameters and LED control must be tightly optimized. Proper biasing, flicker timing, and LED quality directly affect channel integrity and, in turn, depth reconstruction accuracy. Maintaining high temporal resolution and minimizing overlap between LED windows is therefore critical to reliable performance in event-based photometric stereo for tactile sensing.

C. Reconstruction Across Objects and Forces

Building on the experimental setup, we evaluated FlickerTac on eight objects. Four of these (namely, a calibrated sphere, concentric disks, a bumps profile, and a hexagon platform) were used for both quantitative and qualitative analysis. For the sphere ground truth, we followed the analytical procedure in Section IV-A. For the other three objects, depth accuracy was measured at predefined geometric features, averaged over five repeated measurements per feature and three independent contacts per object. The remaining four objects were assessed qualitatively to further test generalization to unseen geometries.

Table II summarizes the quantitative depth reconstruction results for the four reference objects. For each case, the mean absolute error (MAE) and standard deviation were computed across

TABLE II
QUANTITATIVE RESULTS ON FOUR REFERENCE OBJECTS (IN MM)

Object	Contact MAE	Std. Dev.	Off-Contact MAE	% <0.1mm
Sphere	0.311	0.027	0.326	82%
Disks	0.347	0.065	0.381	76%
Bumps	0.394	0.041	0.316	84%
Hexagon	0.328	0.124	0.324	81%

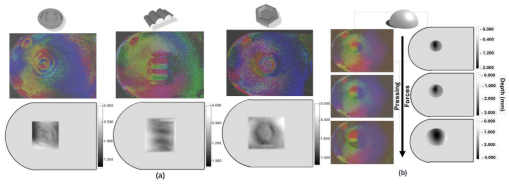


Fig. 7. Qualitative reconstruction on reference objects and sphere indentations under increasing force.

three contacts. The contact region for each object was manually delineated on the cumulative event image, and the reported Contact MAE reflects only pixels within this region. The Off-Contact MAE measures the deviation of background pixels from zero depth (flat plane assumption), reflecting reconstruction artifacts in non-contact areas. Despite background variations, the Off-Contact MAE remains consistently low (0.316–0.381 mm across objects, with per-pixel errors ranging from near-zero to a maximum below 0.9 mm), as over 67% of background pixels are correctly predicted at near-zero depth, indicating effective suppression of spurious depth artifacts outside the contact region. Notably, 76–84% of off-contact pixels exhibit errors below 0.1 mm; the remaining outliers are attributed to the sources discussed in Section V. For the contact region, the calibrated sphere, shown in Fig. 7(b), yielded a reconstruction error of **0.311 ± 0.027 mm**. The concentric disk structure produced a depth MAE of **0.347 ± 0.065 mm**, while the bumps profile and hexagon platform achieved **0.394 ± 0.041 mm** and **0.328 ± 0.124 mm**, respectively. Qualitative reconstruction results for these objects are shown in Fig. 7(a), demonstrating FlickerTac's ability to capture smooth curvature, depth discontinuities, and fine surface variations with high spatial fidelity.

In addition to evaluating geometric variation across objects, we tested FlickerTac's performance under varying contact forces. Using the calibrated sphere, we performed a series of indentations at increasing depths to simulate different pressing intensities. As shown in Fig. 7(b), the sensor consistently produced smooth reconstructions across all force levels, with no degradation in spatial continuity or curvature recovery. This robustness to force variation is particularly valuable for applications involving variable grasp pressure or adaptive manipulation.

To further evaluate generalization, we tested FlickerTac on five additional geometries with varying surface complexity. Fig. 8 shows the tested objects (top row), event-channel inputs (middle row), and the corresponding depth reconstructions (bottom row). The recovered surfaces exhibit strong geometric consistency, clearly defined contours, and detailed local structures, demonstrating the model's robustness across novel shapes. This is made possible by our time-multiplexed flickering illumination scheme, which ensures continuous event generation and structured shading cues, overcoming a longstanding limitation of event-based cameras, which typically require motion to produce signal.

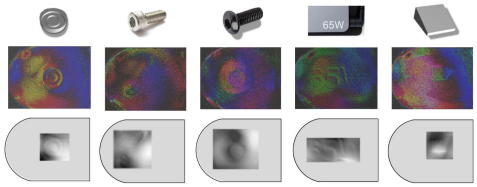


Fig. 8. Depth reconstruction of five unseen objects: views, event channels, and depth maps.

Notably, the tactile skin plays a critical role in depth fidelity. As reported in [36], the DIGIT skin exhibits lower spatial resolution than GelSight skins, potentially limiting fine detail recovery. However, its resilience and ease of integration made it a practical choice for our sensor.

D. Comparison With Representative VBTSs

FlickerTac is the first EVBTS capable of dense 3D reconstruction via photometric stereo. No prior EVBTS system has demonstrated dense surface reconstruction or photometric stereo, limiting comparisons to low temporal resolution frame-based VBTSs despite the inherent differences in sensing modality. This positions FlickerTac as a foundational advancement in bridging asynchronous event sensing with high-resolution tactile geometry estimation. While our evaluation focuses on static 3D reconstruction, it establishes the necessary baseline for dynamic studies, as no prior EVBTS has achieved dense photometric-stereo reconstruction under static contact.

We benchmark FlickerTac against representative, widely used VBTSs selected based on the availability of quantitative depth or surface-normal metrics, including GelSight-based sensors, DenseTact models, and 9DTact, as summarized in Table III. We note that depth MAE is not uniformly reported across all VBTS designs; for example, GelSight sensors are known to achieve high-resolution reconstruction but lack published depth MAE values in comparable settings. Our goal is to demonstrate that FlickerTac provides comparable depth accuracy at substantially higher reconstruction rates. Unlike conventional sensors that rely on RGB imaging and operate at lower frame rates, FlickerTac leverages time-multiplexed flickering illumination and asynchronous event streams to enable low-latency reconstruction at 125 Hz.

The reported FlickerTac results were obtained under the contact reconstruction protocol described in Section IV-A, with angular errors computed from predicted and analytic ground-truth gradients and depth errors evaluated after gradient integration. All error metrics are reported as mean absolute errors (MAE), unless explicitly noted otherwise.

FlickerTac achieves gradient MAE of 0.102/0.074 for G_x/G_y and angular errors of 1.947°/2.597°—closely matching GelSight Wedge (2.248°/2.187°) despite the latter's full RGB intensity access. Depth MAE of 0.311 mm is comparable to DenseTact (0.280 mm) while operating at 125 Hz. Additionally, it achieves a mean depth error of 0.311 mm, comparable to DenseTact (0.280 mm) and lower than DenseTact 2.0 (0.364 mm), while operating at 125 Hz. When the same pipeline is applied to a DIGIT sensor, the depth error reaches 0.266 mm at 60 Hz, confirming that FlickerTac maintains competitive accuracy while offering

TABLE III
COMPARISON OF GRADIENTS AND DEPTH ACROSS SEVERAL VBTSS

Sensor	G_x error	G_y error	θ_x error (deg)	θ_y error (deg)	Depth error (mm)	Rate (Hz)	TS: Tactile Skin	Comments
9DTact [37]	N/A	N/A	N/A	N/A	0.046	90	GelSight like TS	Standard VBTS Not PS; basic lookup table
DenseTact [38]	N/A	N/A	N/A	N/A	0.280	30	Custom TS	Standard VBTS
DenseTact 2.0 [27]	N/A	N/A	N/A	N/A	0.364*	30	Custom TS	Standard VBTS *L2 error
GelSight Wedge [23]	0.043	0.042	2.248	2.187	N/A	30	GelSight TS	Standard VBTS
Digit [22]	0.116	0.065	1.354	2.125	0.266	60	DIGIT TS	Standard VBTS
FlickerTac (Ours)	0.102	0.074	1.947	2.597	0.311	125	DIGIT TS	First EVBTS Dense 3D reconstruction. Outperforms in speed.

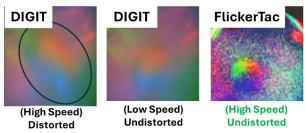


Fig. 9. Rapid sphere rolling causing object distortion due to motion blur.

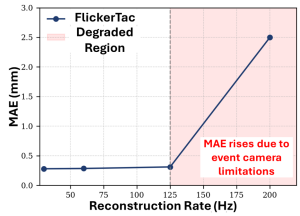


Fig. 10. Reconstruction Rate vs. Depth MAE. Accuracy degrades beyond 125 Hz due to bandwidth limits.

substantially higher reconstruction speed. Although 9DTact reports a smaller error of 0.046 mm, it does so under simplified conditions using lookup tables rather than photometric inference and at lower acquisition rates.

The tactile skin also plays a crucial role in reconstruction accuracy. As noted in [36], the DIGIT skin exhibits lower spatial resolution than GelSight skins, limiting fine detail recovery and directly affecting FlickerTac's results. Nevertheless, we chose the DIGIT skin for its broad community adoption, availability, and resilience to degradation.

Importantly, although we aggregate events within each illumination slot for dense photometric stereo, the event modality's advantages (microsecond latency, high dynamic range, and pixel-level asynchrony) remain the enablers of this performance and provide clear headroom for higher-rate operation as event-camera throughput scales. Enabled by structured temporal illumination, FlickerTac offers a practical path toward real-time and high-speed tactile sensing. The full 3D reconstruction, including gradient regression and Poisson integration, runs at approximately 7 ms per frame, supporting closed-loop feedback in dynamic manipulation tasks. Regarding cost, FlickerTac's cost is dominated by the event camera (same as [12]), while other components match low-cost VBTSs like DIGIT.

E. High-Speed Dynamic Comparison

FlickerTac's primary advantage lies in dynamic tasks where conventional sensors struggle due to motion blur and latency. To highlight this, we conducted a preliminary dynamic test comparing FlickerTac and DIGIT. A sphere was rapidly rolled across the sensing surfaces of both sensors at approximately 20 cm/s. Under rapid lateral contact, DIGIT exhibited severe motion blur and loss of geometry, as seen in the supplementary vide and in Fig. 9. Conversely, FlickerTac's high-temporal channels maintained sharp, continuous contact geometry and consistent shape recovery. Finally, these high-speed dynamic results strongly support future evaluations in continuous slip, vibration sensing, and reactive manipulation.

V. CHALLENGES AND LIMITATIONS

EVBTSs photometric stereo introduces unique challenges not present in VBTSs:

Illumination & Sensor Noise: Phosphor persistence [39], pixel latency jitter [40], and high camera sensitivity to internal skin reflections cause sparse background noise and trailing events (Fig. 6). Together, these stochastic effects produce inevitable frame-to-frame illumination variations.

Event Camera Sensitivity: Unlike frame-based sensors that integrate light, event cameras detect minute brightness changes, including internal reflections and scattered light within the translucent skin, producing sparse background events that appear as "color mixture" in false-color visualizations.

Artifact Mitigation & Poisson Amplification: We mitigate these variations using temporal guard intervals ϵ, 3×3 MLP encoding, and 10% background training sampling. However, residual background noise persists. Crucially, while our predicted normal maps retain strong contact fidelity, the global Poisson integration step amplifies this residual noise into "fog-like" depth artifacts in off-contact regions. Because the MLP cannot fully suppress these stochastic artifacts, quantitative evaluations (Table II) utilize manually demarcated contact masks to isolate valid regions from background deviations. Future research will explore opaque skin layers and regularized, learning-based gradient integration to suppress these depth artifacts.

Modality Trade-off: FlickerTac's 0.311 mm MAE trails DIGIT's 0.266 mm but enables 108% higher bandwidth (125 Hz vs 60 Hz), a favorable speed-accuracy balance. Fig. 10 shows that FlickerTac maintains high accuracy up to 125 Hz; beyond this, temporal separation between LED signals collapses due to readout bandwidth limits (Section III-C), causing inter-channel leakage and degraded gradient inference. Valid rates are further constrained by integer-microsecond LED timing (Section III-C).

VI. Conclusion

We introduced FlickerTac, the first EVBTS achieving dense 3D reconstruction via event-based photometric stereo at 125 Hz. Experimental validation on multiple reference objects demonstrated sub-millimeter depth accuracy, consistent performance across varying contact forces, and effective suppression of background artifacts in off-contact regions. Future work will explore higher-bandwidth event cameras and additional LED channels to push toward kHz-rate tactile sensing.

References

[1] E. Psomopoulou, N. Pestell, F. Papadopoulos, J. Lloyd, Z. Doulgeri, and N. F. Lepora, "A robust controller for stable 3D pinching using tactile sensing," *IEEE Robot. Automat. Lett.*, vol. 6, no. 4, pp. 8150–8157, Oct. 2021.

[2] M. Halwani et al., "A novel vision-based multi-functional sensor for normality and position measurements in precise robotic manufacturing," *Precis. Eng.*, vol. 88, pp. 367–381, Jun. 2024, doi: 10.1016/j.precisioneng.2024.02.015.

[3] I. M. Zaid et al., "Elastomer-based visuotactile sensor for normality of robotic manufacturing systems," *Polymers*, vol. 14, no. 23, 2022, Art. no. 5097. [Online]. Available: https://www.mdpi.com/2073-4360/14/23/5097

[4] A. Solayman et al., "Mechanical behaviour, contact pose estimation, and finite element analysis of vision based tactile sensors fabricated by molding and direct ink writing: A comparative study," *Adv. Eng. Mater.*, vol. 26, no. 22, 2024, Art. no. 2400630.

[5] I. M. Zaid et al., "Virtual prototyping of vision-based tactile sensors design for robotic-assisted precision machining," *Sensors Actuators A: Phys.*, vol. 374, 2024, Art. no. 115469.

[6] M. Salah et al., "NeuTac: Zero-shot sim2real measurement for neuromorphic vision-based tactile sensors," *IEEE Trans. Instrum. Meas.*, vol. 73, 2024, Art. no. 5031315.

[7] G. Cao, J. Jiang, C. Lu, D. F. Gomes, and S. Luo, "Touchroller: A rolling optical tactile sensor for rapid assessment of textures for large surface areas," *Sensors*, vol. 23, no. 5, 2023, Art. no. 2661. [Online]. Available: https://www.mdpi.com/1424-8220/23/5/2661

[8] X. Huang et al., "Neuromorphic vision based contact-level classification in robotic grasping applications," *Sensors*, vol. 20, no. 17, 2020, Art. no. 4724.

[9] A. Rigi, F. B. Naeini, D. Makris, and Y. Zweiri, "A novel event-based incipient slip detection using dynamic active-pixel vision sensor (DAVIS)," *Sensors (Switzerland)*, vol. 18, no. 2, pp. 1–18, 2018.

[10] R. Muthusamy et al., "Neuromorphic eye-in-hand visual servoing," *IEEE Access*, vol. 9, pp. 55853–55870, 2021.

[11] M. Rajkumar, X. Huang, Y. Zweiri, L. Seneviratne, D. Gan, and R. Muthusamy, "Neuromorphic event-based slip detection and suppression in robotic grasping and manipulation," *IEEE Access*, vol. 8, pp. 153364–153384, 2020.

[12] N. Funk, E. Helmut, G. Chalvatzaki, R. Calandra, and J. Peters, "Evetac: An event-based optical tactile sensor for robotic manipulation," *IEEE Trans. Robot.*, vol. 40, pp. 3812–3832, 2024.

[13] H. Sajwani et al., "TactiGraph: An asynchronous graph neural network for contact angle prediction using neuromorphic vision-based tactile sensing," *Sensors*, vol. 23, no. 14, 2023, Art. no. 6451.

[14] B. Ward-Cherrier, N. Pestell, and N. F. Lepora, "NeuroTac: A neuromorphic optical tactile sensor applied to texture recognition," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 2654–2660.

[15] W.-C. Benjamin, J. Conradt, M. G. Catalano, M. Bianchi, and N. F. Lepora, "A miniaturised neuromorphic tactile sensor integrated with an anthropomorphic robot hand," in *Proc. IEEE/RSJ IEEE Int. Conf. Intell. Robots Syst.*, 2020, pp. 9883–9889.

[16] X. Xu, N. F. Lepora, and B. Ward-Cherrier, "A neuromorphic tactile system for reliable braille reading in noisy environments," *IEEE Robot. Automat. Lett.*, vol. 10, no. 6, pp. 5225–5232, Jun. 2025.

[17] F. B. Naeini et al., "A novel dynamic-vision-based approach for tactile sensing applications," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 5, pp. 1881–1893, May 2020.

[18] F. B. Naeini, D. Makris, D. Gan, and Y. Zweiri, "Dynamic-vision-based force measurements using convolutional recurrent neural networks," *Sensors (Switzerland)*, vol. 20, no. 16, 2020, Art. no. 4469.

[19] M. Li, T. Li, and Y. Jiang, "Marker displacement method used in vision-based tactile sensors—from 2-D to 3-D: A review," *IEEE Sensors J.*, vol. 23, no. 8, pp. 8042–8059, Apr. 2023, doi: 10.1109/JSEN.2023.3255861.

[20] M. K. Johnson and E. H. Adelson, "Retrographic sensing for the measurement of surface texture and shape," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2009, pp. 1070–1077.

[21] W. Yuan, S. Dong, and E. H. Adelson, "GelSight: High-resolution robot tactile sensors for estimating geometry and force," *Sensors*, vol. 17, no. 12, 2017.

[22] M. Lambeta et al., "DIGIT: A novel design for a low-cost compact high-resolution tactile sensor with application to in-hand manipulation," *IEEE Robot. Automat. Lett.*, vol. 5, no. 3, pp. 3838–3845, Jul. 2020.

[23] S. Wang, Y. She, B. Romero, and E. H. Adelson, "GelSight wedge: Measuring high-resolution 3D contact geometry with a compact robot finger," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2021, pp. 6468–6475.

[24] S. Dong, W. Yuan, and E. H. Adelson, "Improved GelSight tactile sensor for measuring geometry and slip," in *Proc. IEEE/RSJ IEEE Int. Conf. Intell. Robots Syst.*, 2017, pp. 137–144.

[25] S. Dong, D. Ma, E. Donlon, and A. Rodriguez, "Maintaining grasps within slipping bounds by monitoring incipient slip," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2019, pp. 3818–3824.

[26] I. H. Taylor, S. Dong, and A. Rodriguez, "Gelslim 3.0: High-resolution measurement of shape, force and slip in a compact tactile-sensing finger," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2022, pp. 10781–10787.

[27] W. K. Do, B. Jurewicz, and M. Kennedy, "DenseTact 2.0: Optical tactile sensor for shape and force reconstruction," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2023, pp. 12549–12555.

[28] T. Leroux, S. H. Ieng, and R. Benosman, "Event-based structured light for depth reconstruction using frequency tagged light patterns," 2018, *arXiv:1811.10771*.

[29] H. Rebecq, G. Gallego, E. Mueggler, and D. Scaramuzza, "EMVS: Event-based multi-view stereo—3D reconstruction with an event camera in real-time," *Int. J. Comput. Vis.*, vol. 126, no. 12, pp. 1394–1414, 2018.

[30] B. Yu, J. Ren, J. Han, F. Wang, J. Liang, and B. Shi, "EventPS: Real-time photometric stereo using an event camera," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 9602–9611.

[31] R. J. Woodham, "Photometric method for determining surface orientation from multiple images," *Opt. Eng.*, vol. 19, no. 1, pp. 139–144, 1980.

[32] M. K. Johnson, F. Cole, A. Raj, and E. H. Adelson, "Microgeometry capture using an elastomeric sensor," *ACM Trans. Graph.*, vol. 30, no. 4, pp. 1–8, Jul. 2011, doi: 10.1145/2010324.1964941.

[33] iniVation AG, "Understanding the performance of neuromorphic event-based vision sensors," iniVation AG, Zurich, Switzerland, White Paper, May 2020. Accessed: Jun. 14, 2025. [Online]. Available: https://inivation.com/wp-content/uploads/2020/06/iniVation_White_Paper_Performance_of_Event_Cameras.pdf

[34] A. Ajoudani et al., "Reflex control of the Pisa/IIT SoftHand during object slippage," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2016, pp. 1972–1979.

[35] Q. Li, O. Kroemer, Z. Su, F. F. Veiga, M. Kaboli, and H. J. Ritter, "A review of tactile information: Perception and action through touch," *IEEE Trans. Robot.*, vol. 36, no. 6, pp. 1619–1634, Dec. 2020.

[36] H. Gupta, Y. Mo, S. Jin, and W. Yuan, "Sensor-invariant tactile representation," in *Proc. 13th Int. Conf. Learn. Representations*, 2025, pp. 74682–74705.

[37] C. Lin, H. Zhang, J. Xu, L. Wu, and H. Xu, "9DTact: A compact vision-based tactile sensor for accurate 3D shape reconstruction and generalizable 6D force estimation," *IEEE Robot. Automat. Lett.*, vol. 9, no. 2, pp. 923–930, Feb. 2024.

[38] W. K. Do and M. Kennedy, "Densetact: Optical tactile sensor for dense shape reconstruction," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2022, pp. 6188–6194.

[39] H. Lin et al., "Phosphor-in-glass for high-powered remote-type white AC-LED," *ACS Appl. Mater. Interfaces*, vol. 6, no. 23, pp. 21264–21269, 2014.

[40] R. W. Baldwin, M. Almatrafi, V. Asari, and K. Hirakawa, "Event probability mask (EPM) and event denoising convolutional neural network (EDnCNN) for neuromorphic cameras," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 1698–1707.